Teacher answer key: Primary 5, Semester 2 (DRAFT)

DRAFT: this answer key is for owner review. Do not give it to pupils. Contact: pypteltd@gmail.com (business email for now, to be changed later)

This key lists the answers and the result of the check on every page. You do not need it for routine marking: a pupil who has a working check has the right answers. To spot-check, compare the check strip with the line "Check result" below.

How the topic tags work

Each entry names the syllabus outcomes the puzzle practises, using the outcome lines and numbers of the Primary 5 (Standard) content in the Mathematics Syllabus, Primary (2021 edition, updated October 2025, pages 41 and 42), copied as printed there. Everything else in each entry is our own wording. The references are for your planning only and are not printed on the pupil pages. "Problem solving" marks a page whose questions come from the kinds found in school tests rather than from one numbered outcome.

Pages in this booklet (Terms 3 and 4)

Strand	Pages
Number and Algebra - Percentage	S2-01, S2-02, S2-03, S2-04, S2-05
Number and Algebra - Rate	S2-06, S2-07
Measurement and Geometry - Area and Volume	S2-08, S2-09, S2-10, S2-11, S2-12, S2-13, S2-14, S2-15
Measurement and Geometry - Geometry	S2-16, S2-17, S2-18, S2-19, S2-20

The page code in the footer, for example S2-07 (P5-27), gives the page in this booklet and, in brackets, its number in the full set of 40.

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S2-01 (P5-21) How Many Percent? 1 star

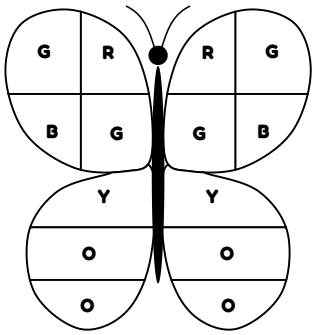
Topic: Percentage: a part of a whole as a percentage

Syllabus outcomes: Percentage 1.1, Percentage 1.2. Notes on the tag: expressing a part of a whole as a percentage; the sign %
Puzzle type: Code-breaker (percentages). How the pupil checks: Code word (6 letters).
Answers: 1) 10%; 2) 40%; 3) 5%; 4) 30%; 5) 75%; 6) 98%
Check result: Code word: BRANDS
Teacher note: Six questions: four bars divided into equal parts with some parts shaded, and two stories with a part and a whole. The pupil writes the part of the whole as a percentage. The decoy letters stand for the usual slips: giving the number of shaded parts, giving the percentage not shaded, or giving the number of parts left over. A box left blank gives no letter and the check cannot see it.

S2-02 (P5-22) Colour the Butterfly 1 star

Topic: Percentage: a part of a whole, a percentage of a quantity, a price after a discount or a percentage added

Syllabus outcomes: Percentage 1.1, Percentage 1.2, Percentage 1.3, Percentage 1.4. Notes on the tag: a percentage part of a whole, a part of a whole as a percentage, discount and GST (percentage)
Puzzle type: Colour by code (pupils colour with their own crayons). How the pupil checks: Colour key and tally: every answer must be on the key, the number of parts of each colour must match the box, and the parts in the same place on the left and on the right match.
Answers: Colour key: 12 = yellow, 20 = red, 25 = blue, 40 = green, 60 = orange
Check result: Tally: yellow 2, red 2, blue 2, green 4, orange 4; the parts in the same place on the left and on the right match.



Teacher note: Every part asks for a percentage of a quantity, a part of a whole written as a percentage, a price after a discount, or a price after adding a percentage such as GST. Pupils colour the black-and-white outline with their own crayons (or, for a pencil-only copy, write Y, R, B, G or O in each part: the tally and the matching parts can be checked the same way). Check strip: count the parts of each colour (the tally is printed); the parts in the same place on the left and on the right match.

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S2-03 (P5-23) Sale Prices 2 stars

Topic: Percentage: finding the discount and the sale price

Syllabus outcomes: Percentage 1.4, Percentage 1.3. Notes on the tag: finding discount and the price after discount
Puzzle type: Ordering (lettered cards). How the pupil checks: Letters of the cards in the order asked spell a word (6 letters).
Answers: Order (smallest sale price first): Price: $30 Discount: 50% -> $15, Price: $30 Discount: 25% -> $22.50, Price: $30 Discount: 15% -> $25.50, Price: $100 Discount: 40% -> $60, Price: $250 Discount: 20% -> $200, Price: $250 Discount: 10% -> $225
Check result: Word: LAUGHS
Teacher note: Six price tags, each with an original price and a percentage discount. The pupil finds the discount, takes it off the price and puts the six sale prices in order from smallest to greatest. The cards are not in order of sale price. The letters, read in that order, spell a word with exactly one arrangement that is a word, so no wrong order spells a word. A box left blank gives no letter and the check cannot see it.

S2-04 (P5-24) GST Match 2 stars

Topic: Percentage: GST

Syllabus outcomes: Percentage 1.4, Percentage 1.3. Notes on the tag: finding GST and the price with GST
Puzzle type: Matching (lettered cards). How the pupil checks: Letters of the matching cards, tile 1 to 6, spell a word.
Answers: F: Before GST: $45 -> With GST: $49.05; U: Before GST: $55 -> With GST: $59.95; S: Before GST: $60 -> With GST: $65.40; I: Before GST: $155 -> With GST: $168.95; O: Before GST: $160 -> With GST: $174.40; N: Before GST: $165 -> With GST: $179.85
Check result: Word: FUSION
Teacher note: Six cards with a price before GST (multiples of $5) and six tiles with the price with 9% GST, in order of size. The pupil finds 9% of the price, adds it, and matches the card with the tile. The letters, read from tile 1 to tile 6, spell a word with exactly one arrangement that is a word, so no wrong matching spells a word. GST is 9%; the page states the rate. A box left blank gives no letter and the check cannot see it.

S2-05 (P5-25) Annual Interest 2 stars

Topic: Percentage: annual interest

Syllabus outcomes: Percentage 1.4, Percentage 1.3 (problem solving). Notes on the tag: finding annual interest
Puzzle type: Code-breaker (interest in money stories). How the pupil checks: Code word (6 letters).
Answers: 1) $1236; 2) $60; 3) $20; 4) $40; 5) $206; 6) $2050
Check result: Code word: COSTLY
Teacher note: Six savings stories with a percentage interest paid at the end of the year, in whole dollars. The interest is a percentage of the first amount saved (the page says so); the pupil finds the interest for one year or the money in the bank after one year (every story is one year, as in the school papers; the syllabus line says only annual interest). The decoy letters stand for the usual slips: giving the interest when the total is asked, the total when the interest is asked, or the interest for two years. Interest on interest (compound interest) is not used. A box left blank gives no letter and the check cannot see it.

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S2-06 (P5-26) Rate Cards 1 star

Topic: Rate: finding the rate, the total or the number of units

Syllabus outcomes: Rate 1.1, Rate 1.2 (problem solving). Notes on the tag: rate as the amount of one quantity per unit of another quantity; finding the rate, the total or the number of units
Puzzle type: Lettered choices (rate). How the pupil checks: Word from the letters of the right choices (6 letters).
Answers: 1) 5 ℓ; 2) $9.60; 3) 6 minutes; 4) 12 buns; 5) 48 kg; 6) 3 m
Check result: Word: TYPING
Teacher note: Six one-step rate questions: two ask for the rate (an amount for one unit), two for the total at a given rate and two for the number of units. The wrong choices are the usual slips (dividing the wrong way round, multiplying instead of dividing, adding or subtracting the two numbers). Each choice has a letter and the right choices spell the word. Every other way of choosing was tested against a very large word list. A box left blank gives no letter and the check cannot see it.

S2-07 (P5-27) Rate Stories 2 stars

Topic: Rate: two-step problems

Syllabus outcomes: Rate 1.1, Rate 1.2 (problem solving). Notes on the tag: rate: finding the rate, then the total or the number of units
Puzzle type: Code-breaker (rate stories). How the pupil checks: Code word (6 letters).
Answers: 1) 154 ℓ; 2) 20 hours; 3) 55 dollars; 4) 48 bikes; 5) 18 hours; 6) 121 dollars
Check result: Code word: DIGEST
Teacher note: Six two-step rate stories: the pupil works out the answer from the rate in the story: a total for a stated number of units, or the number of units for a stated total. All answers are whole numbers. The decoy letters stand for the usual slips: the two numbers turned upside down, or only the rate given. A box left blank gives no letter and the check cannot see it.

S2-08 (P5-28) Height Hunt 1 star

Topic: Area of a triangle: the base and its height

Syllabus outcomes: Area of Triangle 1.1. Notes on the tag: area of a triangle: base and height (the height is at a right angle to the base)
Puzzle type: Lettered choices (figures). How the pupil checks: Word from the letters of the right choices (6 letters).
Answers: 1) HF (base GU); 2) KG (base JM); 3) NU (base LV); 4) JT (base WQ); 5) SQ (base FE); 6) PG (base QH)
Check result: Word: HOURLY
Teacher note: Six triangles, each with a named base. One dashed line from the opposite vertex meets the base line at a right angle (it is the height); a second dashed line and a side of the triangle do not. 1 of the six bases are tilted and 0 of the six heights fall outside the triangle (the base is then extended with a dotted line). The page says the figures are drawn to scale, so a pupil can check a right angle with the corner of a sheet of paper or a set square. This is the only page of the two booklets that relies on the drawing. The wrong choices are the slanted line and a side. Each choice has a letter and the right choices spell the word. Every other way of choosing was tested against a very large word list. A box left blank gives no letter and the check cannot see it.

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S2-09 (P5-29) Triangle Areas 1 star

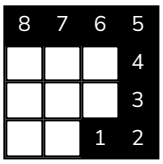
Topic: Area of a triangle: base, height and area

Syllabus outcomes: Area of Triangle 1.1, Area of Triangle 1.2. Notes on the tag: area of a triangle from its base and height
Puzzle type: Code-breaker (areas of triangles). How the pupil checks: Code word (6 letters).
Answers: 1) 35 cm² (base 10, height 7); 2) 27 cm² (base 9, height 6); 3) 25 cm² (base 10, height 5); 4) 18 cm² (base 6, height 6); 5) 16 cm² (base 8, height 4); 6) 77 cm² (base 14, height 11)
Check result: Code word: BIRTHS
Teacher note: Six triangles with the base and the height (dashed, at a right angle to the base) labelled in cm; 0 of the six figures have a length beside a slanted side that is not needed, and 0 have the height outside the triangle (the base is then extended with a dotted line). The decoy letters stand for the usual slips: forgetting to halve, adding the base and the height, or using the slanted side instead of the height. A box left blank gives no letter and the check cannot see it. The figures are sketches and the page does not rely on the drawing: the pupil uses the printed numbers.

S2-10 (P5-30) Triangle Route 2 stars

Topic: Area of a triangle: base, height and area

Syllabus outcomes: Area of Triangle 1.2. Notes on the tag: area of a triangle from its base and height
Puzzle type: Maze / path. How the pupil checks: Route: the line must reach FINISH through the right number of boxes, every box following the rule.
Answers: Route (row, column from top-left): (4,3) 12 x 5 / 2 = 30 -> (4,4) 5 x 12 / 2 = 30 -> (3,4) 6 x 10 / 2 = 30 -> (2,4) 10 x 6 / 2 = 30 -> (1,4) 6 x 10 / 2 = 30 -> (1,3) 6 x 10 / 2 = 30 -> (1,2) 10 x 6 / 2 = 30 -> (1,1) 5 x 12 / 2 = 30
Check result: Route reaches FINISH through 8 boxes; every box on it follows the rule.



Teacher note: A 4 by 4 grid of sixteen triangles, each with its base and its height (dashed) labelled in cm; 1 of the sixteen have the height outside the triangle (the base is then extended with a dotted line). The boxes on the route have area 30 cm². The boxes off the route have a product base x height that gives the area without halving, or an area a little more or less than 30 cm²; some extra boxes that do have area 30 cm² are dead ends. Counting the boxes shows a line that takes a shortcut. The figures are sketches and the page does not rely on the drawing: the pupil uses the printed numbers.

S2-11 (P5-31) Composite Figures 2 stars

Topic: Area of composite figures: rectangles and triangles

Syllabus outcomes: Area of Triangle 1.2, Area of Triangle 1.3. Notes on the tag: area of composite figures made of rectangles and triangles
Puzzle type: Code-breaker (composite areas). How the pupil checks: Code word (6 letters).
Answers: 1) 48 cm² (house); 2) 36 cm² (house); 3) 18 cm² (notch); 4) 39 cm² (side); 5) 34 cm² (side); 6) 32 cm² (notch)
Check result: Code word: THIRDS
Teacher note: Six figures: a rectangle with a triangle on top (two figures), a rectangle with a triangular notch cut from its top edge (two figures) and a rectangle with a right-angled triangle joined to one side (two figures). The pupil finds the area of the rectangle and of the triangle and adds or subtracts. The decoy letters stand for the usual slips: forgetting to halve the triangle, or adding where the notch must be taken away. Dashed lines are not edges of the figure. The figures are sketches and the page does not rely on the drawing: the pupil uses the printed numbers. A box left blank gives no letter and the check cannot see it.

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S2-12 (P5-32) Count the Cubes 1 star

Topic: Volume: building cuboids with unit cubes

Syllabus outcomes: Volume 2.1, Volume 2.3, Volume 2.4. Notes on the tag: building cubes and cuboids with unit cubes; reading drawings of cuboids on an isometric grid; volume of a cuboid
Puzzle type: Code-breaker (cubes in a cuboid). How the pupil checks: Code word (6 letters).
Answers: 1) 3 x 3 x 2 = 18; 2) 2 x 2 x 2 = 8; 3) 2 x 1 x 2 = 4; 4) 3 x 2 x 1 = 6; 5) 4 x 3 x 1 = 12; 6) 4 x 2 x 2 = 16
Check result: Code word: JOINED
Teacher note: Six cuboids built of unit cubes, drawn from above in isometric view with every unit face visible on the top and two sides. The pupil counts the cubes in one layer and the layers, or multiplies the three edges. The decoy letters stand for the usual slips: counting only the faces that can be seen, adding the three edges, or leaving a layer out. Drawing a cuboid on an isometric grid cannot be checked on a printed page: this page practises reading such a drawing. A box left blank gives no letter and the check cannot see it.

S2-13 (P5-33) Cuboid Volumes 1 star

Topic: Volume of a cube and a cuboid; cm^3 and m^3

Syllabus outcomes: Volume 2.4, Volume 2.2. Notes on the tag: volume of a cube or cuboid in cubic units (cm^3 and m^3)
Puzzle type: Code-breaker (volumes). How the pupil checks: Code word (6 letters).
Answers: 1) 567 cm^3; 2) 126 cm^3; 3) 252 m^3; 4) 616 cm^3; 5) 27 cm^3; 6) 196 m^3
Check result: Code word: SURELY
Teacher note: Six solids with their edges labelled: four cuboids and one cube in cm (the cube has the same edge on all three labels) and, on two figures, in m. The pupil multiplies the three edges; the unit of the answer is cm^3 or m^3 (the page prints it; there is no conversion between them). The decoy letters stand for the usual slips: adding the edges, the area of a face, or leaving one edge out. The figures are sketches and the page does not rely on the drawing: the pupil uses the printed numbers. A box left blank gives no letter and the check cannot see it.

S2-14 (P5-34) Tank Match 2 stars

Topic: Volume: litres and cubic centimetres

Syllabus outcomes: Volume 2.4, Volume 2.6. Notes on the tag: volume of a cuboid; the relationship between litres and cm^3 (1 ℓ = 1000 cm^3)
Puzzle type: Matching (lettered cards). How the pupil checks: Letters of the matching cards, tile 1 to 6, spell a word.
Answers: T: 25 x 40 x 12 = 12000 cm^3 = 12 ℓ; A: 25 x 20 x 30 = 15000 cm^3 = 15 ℓ; L: 80 x 20 x 12 = 19200 cm^3 = 19.2 ℓ; K: 25 x 20 x 40 = 20000 cm^3 = 20 ℓ; E: 50 x 15 x 30 = 22500 cm^3 = 22.5 ℓ; D: 80 x 40 x 15 = 48000 cm^3 = 48 ℓ
Check result: Word: TALKED
Teacher note: Six tanks drawn with their length, width and height in cm, and six tiles with a capacity in litres, in order of size. The pupil multiplies the three edges to find the volume in cm^3 and changes it to litres (1 ℓ = 1000 cm^3). The letters, read from tile 1 to tile 6, spell a word with exactly one arrangement that is a word, so no wrong matching spells a word. The figures are sketches and the page does not rely on the drawing: the pupil uses the printed numbers. A box left blank gives no letter and the check cannot see it.

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S2-15 (P5-35) Tank Tales 2 stars

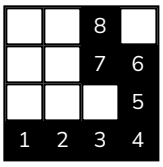
Topic: Volume of liquid in a rectangular tank; litres and cm³

Syllabus outcomes: Volume 2.5, Volume 2.6, Volume 2.4 (problem solving). Notes on the tag: volume of liquid in a rectangular tank; the relationship between ℓ, mℓ and cm³
Puzzle type: Code-breaker (tank stories). How the pupil checks: Code word (6 letters).
Answers: 1) 900 mℓ; 2) 8 ℓ; 3) 115 ℓ; 4) 75 ℓ; 5) 35 ℓ; 6) 50 ℓ
Check result: Code word: FORMAT
Teacher note: Six tank stories: the capacity of a full tank, the water in a tank filled to a stated depth, the water still needed to fill a tank, the capacity of a small box in mℓ, a cube-shaped tank, and the difference between two tanks. Every step goes forward from the lengths to the volume; no story asks for a missing length or for a water level from a volume. The decoy letters stand for the usual slips: the answer left in cm³ or changed by the wrong power of ten. A box left blank gives no letter and the check cannot see it.

S2-16 (P5-36) Angle Route 1 star

Topic: Geometry: angles on a straight line, at a point and vertically opposite angles

Syllabus outcomes: Geometry 1.1, Geometry 1.2, Geometry 1.3, Geometry 1.4. Notes on the tag: angles on a straight line, angles at a point, vertically opposite angles, finding unknown angles
Puzzle type: Maze / path. How the pupil checks: Route: the line must reach FINISH through the right number of boxes, every box following the rule.
Answers: Route (row, column from top-left): (4,1) ? = 70° -> (4,2) ? = 70° -> (4,3) ? = 70° -> (4,4) ? = 70° -> (3,4) ? = 70° -> (2,4) ? = 70° -> (2,3) ? = 70° -> (1,3) ? = 70°
Check result: Route reaches FINISH through 8 boxes; every box on it follows the rule.



Teacher note: A 4 by 4 grid of sixteen angle figures: angles on a straight line (they make a straight angle), angles at a point (a full turn) and two straight lines crossing (vertically opposite angles are equal; neighbouring angles make a straight angle). The angle marked ? is 70° on the route. The boxes off the route give an answer a little more or less than 70°, or the angle next to it; some extra boxes that do give 70° are dead ends. Counting the boxes shows a line that takes a shortcut. Figures are not drawn to scale.

S2-17 (P5-37) Find the Angle 1 star

Topic: Geometry: finding unknown angles

Syllabus outcomes: Geometry 1.1, Geometry 1.2, Geometry 1.3, Geometry 1.4. Notes on the tag: angles on a straight line, angles at a point, vertically opposite angles, finding unknown angles
Puzzle type: Lettered choices (figures). How the pupil checks: Word from the letters of the right choices (6 letters).
Answers: 1) 60°; 2) 70°; 3) 115°; 4) 40°; 5) 50°; 6) 145°
Check result: Word: BACKED
Teacher note: Six figures, two of each kind: angles on a straight line (a straight angle), angles at a point (a full turn) and two crossing lines (vertically opposite angles equal, neighbours make a straight angle). The wrong choices are the usual slips: the angle next to the right one, a full turn used on a straight line, or the sum of the given angles. Each choice has a letter and the right choices spell the word. Every other way of choosing was tested against a very large word list. Figures are not drawn to scale. A box left blank gives no letter and the check cannot see it.

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S2-18 (P5-38) Triangle Angles 1 star

Topic: Geometry: angle sum of a triangle; isosceles, equilateral and right-angled triangles

Syllabus outcomes: Geometry 2.1, Geometry 2.2, Geometry 2.3, Geometry 1.4. Notes on the tag: angle sum of a triangle; properties of isosceles, equilateral and right-angled triangles; finding unknown angles
Puzzle type: Code-breaker (angles in triangles). How the pupil checks: Code word (6 letters).
Answers: 1) 56° (sum); 2) 100° (base); 3) 48° (apex); 4) 60° (equi); 5) 91° (sum); 6) 28° (right)
Check result: Code word: LIFTED
Teacher note: Six triangles: two with two angles given (the three angles of a triangle make a straight angle), an isosceles triangle with the angle at the top given, an isosceles triangle with a base angle given, an equilateral triangle and a right-angled triangle with one other angle given. Equal sides carry tick marks. The decoy letters stand for the usual slips: giving a straight angle minus one angle only, adding the angles, or not halving the base angles. The page does not rely on the drawing: the pupil should use the angle facts, not a protractor. A box left blank gives no letter and the check cannot see it.

S2-19 (P5-39) Shape Statements 2 stars

Topic: Geometry: parallelogram, rhombus and trapezium

Syllabus outcomes: Geometry 3.1, Geometry 3.2. Notes on the tag: properties of a parallelogram, rhombus and trapezium; finding unknown angles
Puzzle type: True or false table (three choices). How the pupil checks: Word from the letters of the right choices (6 letters).
Answers: 1) F (KLMN is a trapezium. KL is parallel to NM. KN and LM are the other two sides. Given: ∠N = 79°. Claim: ∠K = 79°.); 2) N (EFGH is a trapezium. EF is parallel to HG. EH and FG are the other two sides. Given: ∠H = 54°. Claim: ∠G = 54°.); 3) T (ABCD is a rhombus. Given: ∠D = 79°. Claim: ∠A = 101°.); 4) T (TUVW is a trapezium. TU is parallel to WV. TW and UV are the other two sides. Given: ∠U = 61°. Claim: ∠V = 119°.); 5) F (WXYZ is a parallelogram. Given: ∠W = 62°. Claim: ∠X = 62°.); 6) T (PQRS is a parallelogram. Given: ∠Q = 77°. Claim: ∠S = 77°.)
Check result: Word: MATRIX
Teacher note: Six rows, each with a shape, one given angle and a claim about a second angle. A trapezium here has at least one pair of parallel sides; the rows marked N compare the two angles on the same parallel side, so the answer is the same if a trapezium is taken to have exactly one pair. The pupil decides whether the claim must be true, must be false, or cannot be told from the given facts. In a parallelogram or rhombus opposite angles are equal and neighbouring angles make a straight angle; in a trapezium the two angles on the same non-parallel side make a straight angle (the page names the parallel sides and the other two sides), and two angles that are not on the same non-parallel side cannot be found from each other, so the answer is N. Each choice (T, F, N) has a letter and the right choices spell the word. Every other way of choosing was tested against a very large word list. A box left blank gives no letter and the check cannot see it.

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S2-20 (P5-40) Angle Chains 2 stars

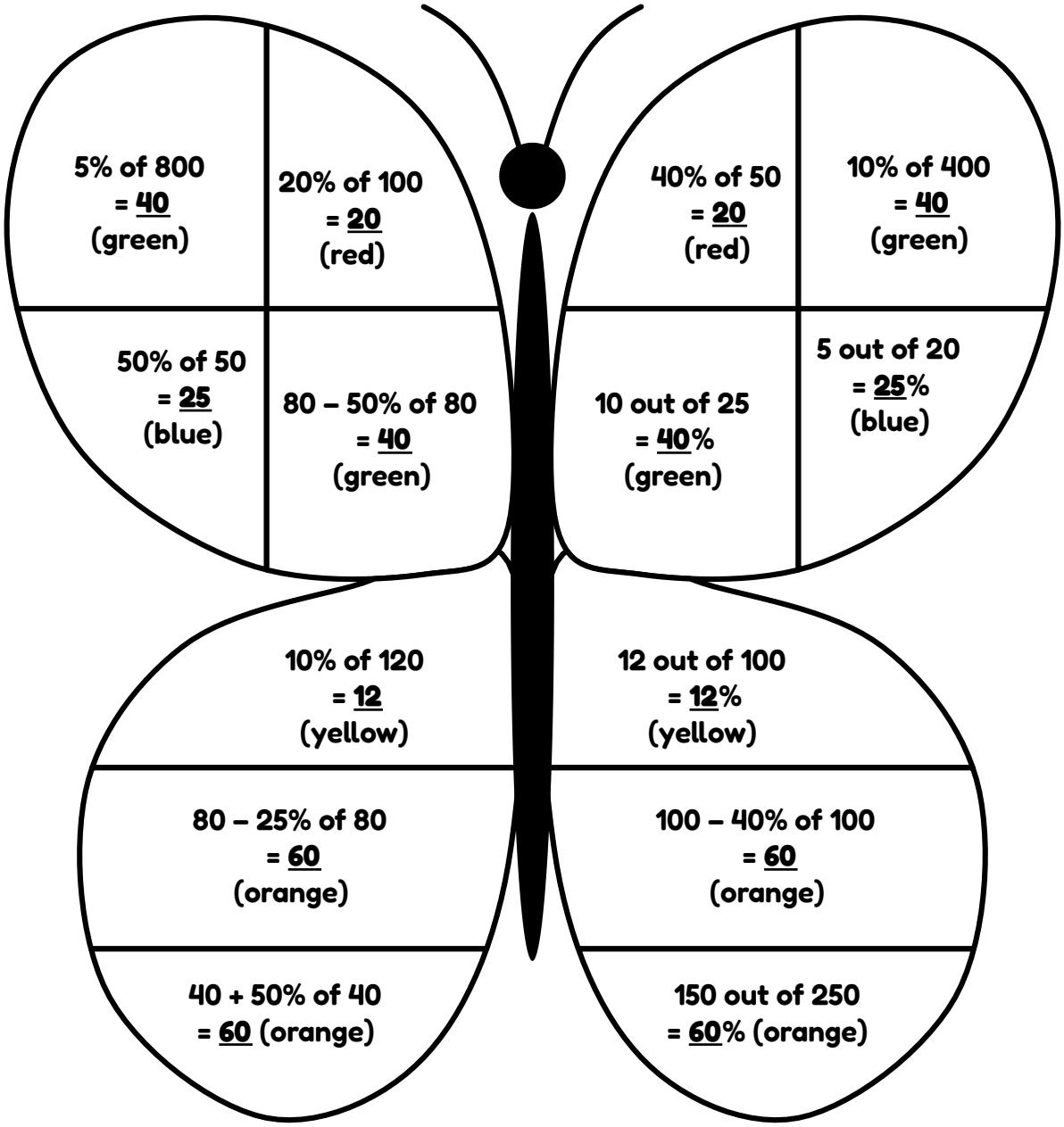
Topic: Geometry: unknown angles in two steps

Syllabus outcomes: Geometry 1.4, Geometry 2.2, Geometry 2.3, Geometry 1.1, Geometry 1.3, Geometry 2.1 (problem solving). Notes on the tag: finding unknown angles in two steps: angle sum of a triangle with angles on a straight line or vertically opposite angles
Puzzle type: Code-breaker (angle chains). How the pupil checks: Code word (6 letters).
Answers: 1) 128° (ext); 2) 119° (isoext); 3) 112° (ext); 4) 53° (ext2); 5) 84° (ext2); 6) 130° (isoext)
Check result: Code word: DEPTHS
Teacher note: Six two-step figures, two of each kind: a triangle with two angles given and one side extended (find the third angle, then the angle on the straight line); an isosceles triangle with the top angle given and the base extended (find the base angle, then the angle on the straight line); and a triangle on a straight line with the two outside angles given (find the two inside angles on the line, then the third angle of the triangle). The decoy letters stand for the usual slips: the angle inside the triangle instead of the angle on the line, or stopping after the first step. Figures are not drawn to scale. A box left blank gives no letter and the check cannot see it.

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Answers: S2-02 (P5-22) Colour the Butterfly

Every part shows the question, the answer underlined and the colour in brackets. Pupils colour their own copy. Colour key: 12 = yellow, 20 = red, 25 = blue, 40 = green, 60 = orange.



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